COMMON RAIL REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

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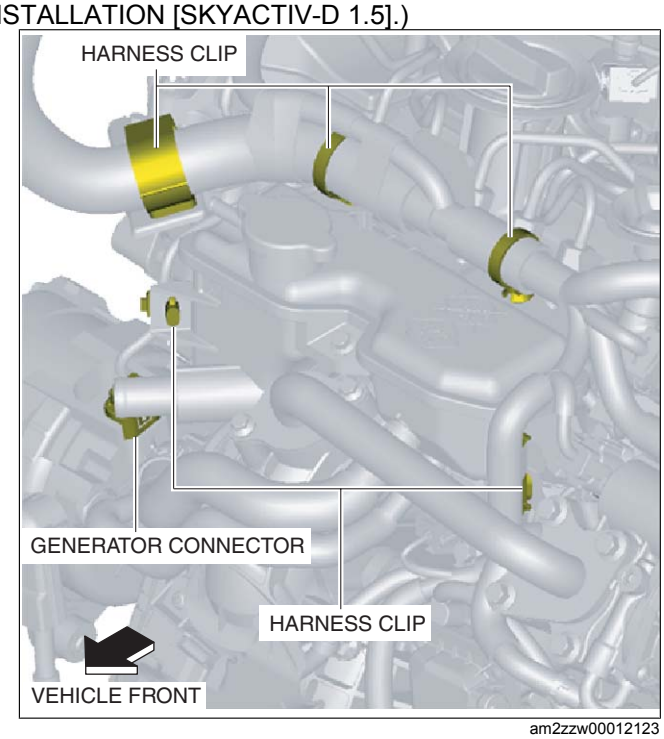
Warning

- **Fuel is highly flammable and dangerous. Fuel line spills and leakage can cause serious injury or death, and damage to equipment. If fuel comes into contact with the skin and eyes it will cause irritation. To prevent this, always perform the "Fuel Line Safety Procedure" referring to the "BEFORE REPAIR PROCEDURE".**
- **A person charged with static electricity could cause a fire or explosion, resulting in death or serious injury. Before servicing the fuel system, make sure to discharge static electricity by touching a vehicle.**

Caution

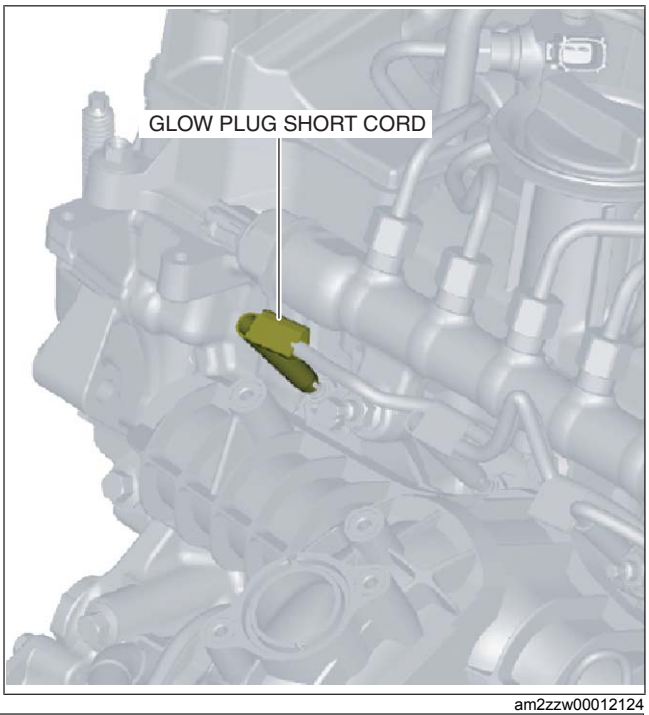
- **If parts are exposed to the fuel, their function may be decreased or a fire may occur. Spread a rag to absorb any leaked fuel.**
- **Do not let foreign matter get into the injection pipe to prevent clogging. In addition, always tighten the injection pipe to the specified torque.**

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Complete the "BEFORE SERVICE PRECAUTION". (See BEFORE SERVICE PRECAUTION [SKYACTIV-D 1.5].)
4. Remove the EGR pipe. (See EGR PIPE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5. Remove the wiring harness clips and the generator connector shown in the figure.
6. Set the water-cooled charge air cooler reserve tank aside so that it does not interfere with the servicing. (See WATER-COOLED CHARGE AIR COOLER RESERVE TANK REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
7. Remove the air Inlet pipe. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
8. Remove the injection pipe (fuel injector side) and the injection pipe (supply pump side). (See INJECTION PIPE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

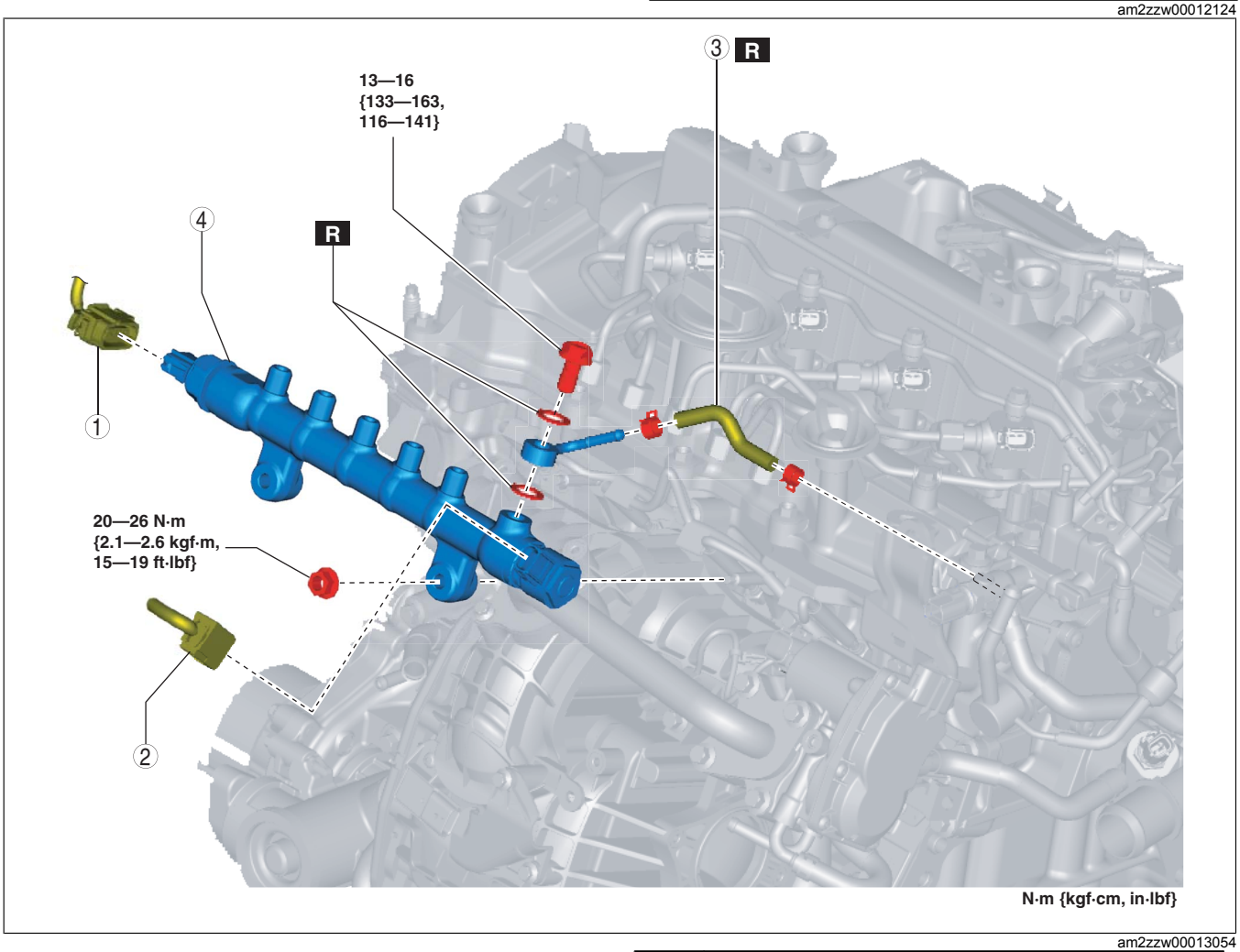


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9. Disconnect the glow plug short cord shown in the figure.
10. Remove in the order shown in the figure.
11. Install in the reverse order of removal.
12. Complete the “AFTER SERVICE PRECAUTION”. (See AFTER SERVICE PRECAUTION [SKYACTIV-D 1.5].)



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1	Fuel pressure sensor connector
2	Fuel pressure relief valve connector
3	Fuel return hose (See Fuel Return Hose Installation Note.)

4	Common rail (See Common Rail Installation Note.)

Common Rail Installation Note

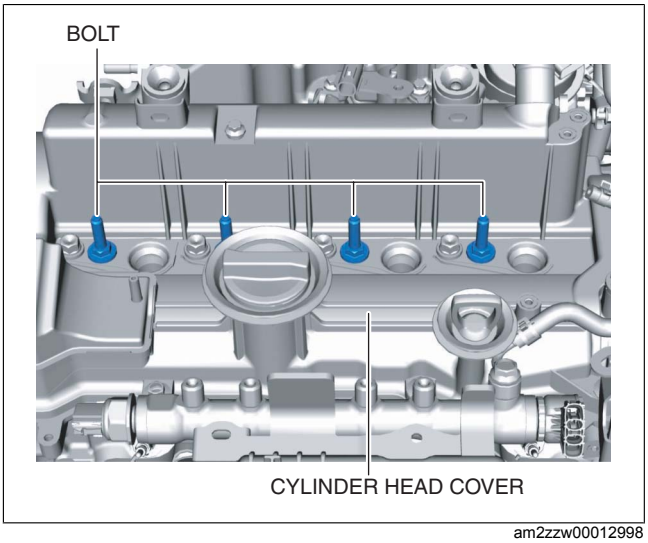
1. Remove the fuel injector bracket. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
2. Remove the fuel return pipe. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Tighten the bolts shown in the figure to the specified torque.

Caution

- **The bolts shown in the figure may loosen when removing the fuel injector bracket because the nuts assembled to the bolts are removed. Tighten the bolts shown in the figure to the specified torque before installing the fuel injector bracket.**

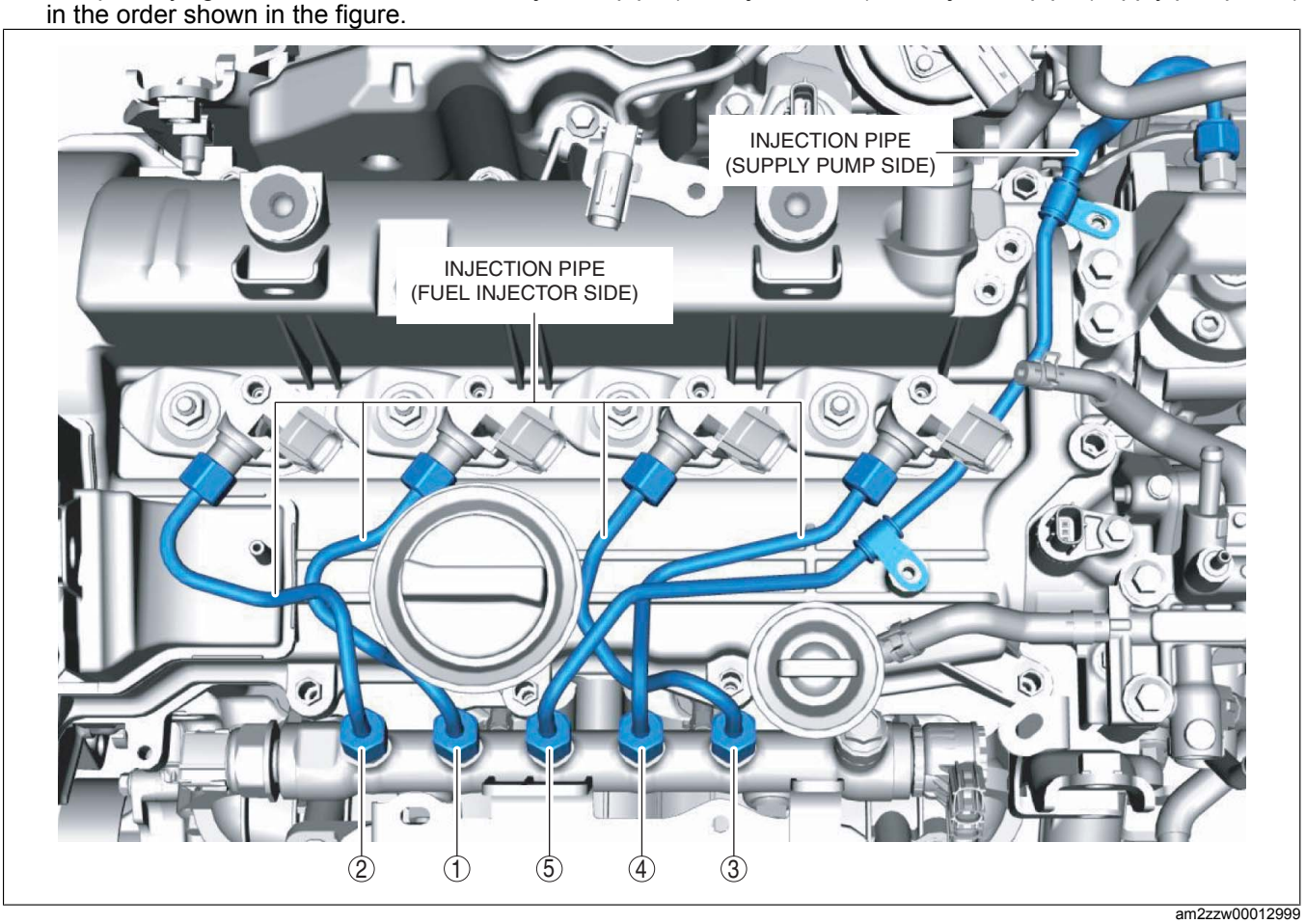
Tightening torque
15—20 N·m {1.6—2.0 kgf·cm, 12—14 in·lbf}

4. Temporarily install the fuel injector bracket.
5. Temporarily tighten the common rail installation nut.



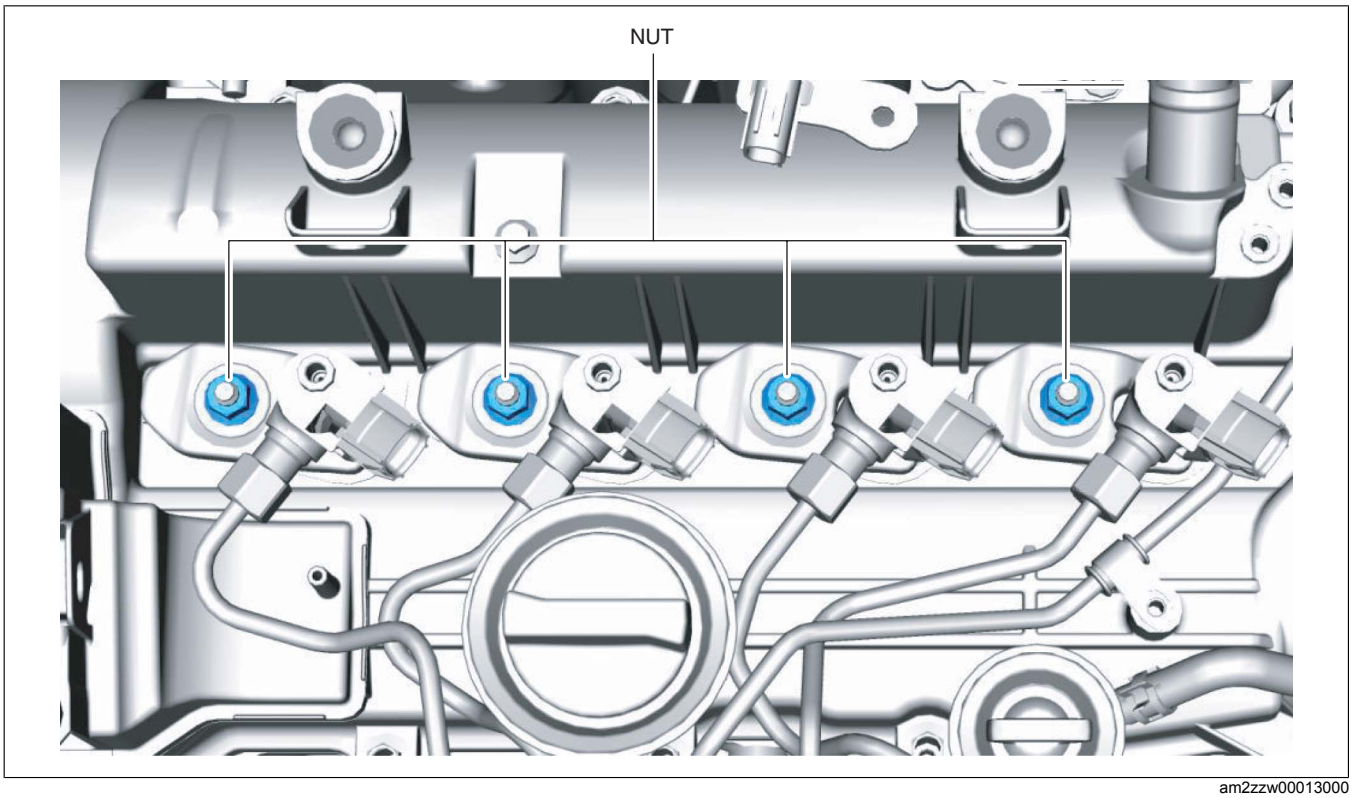
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6. Temporarily tighten both ends of the new injection pipe (fuel injector side) and injection pipe (supply pump side) in the order shown in the figure.



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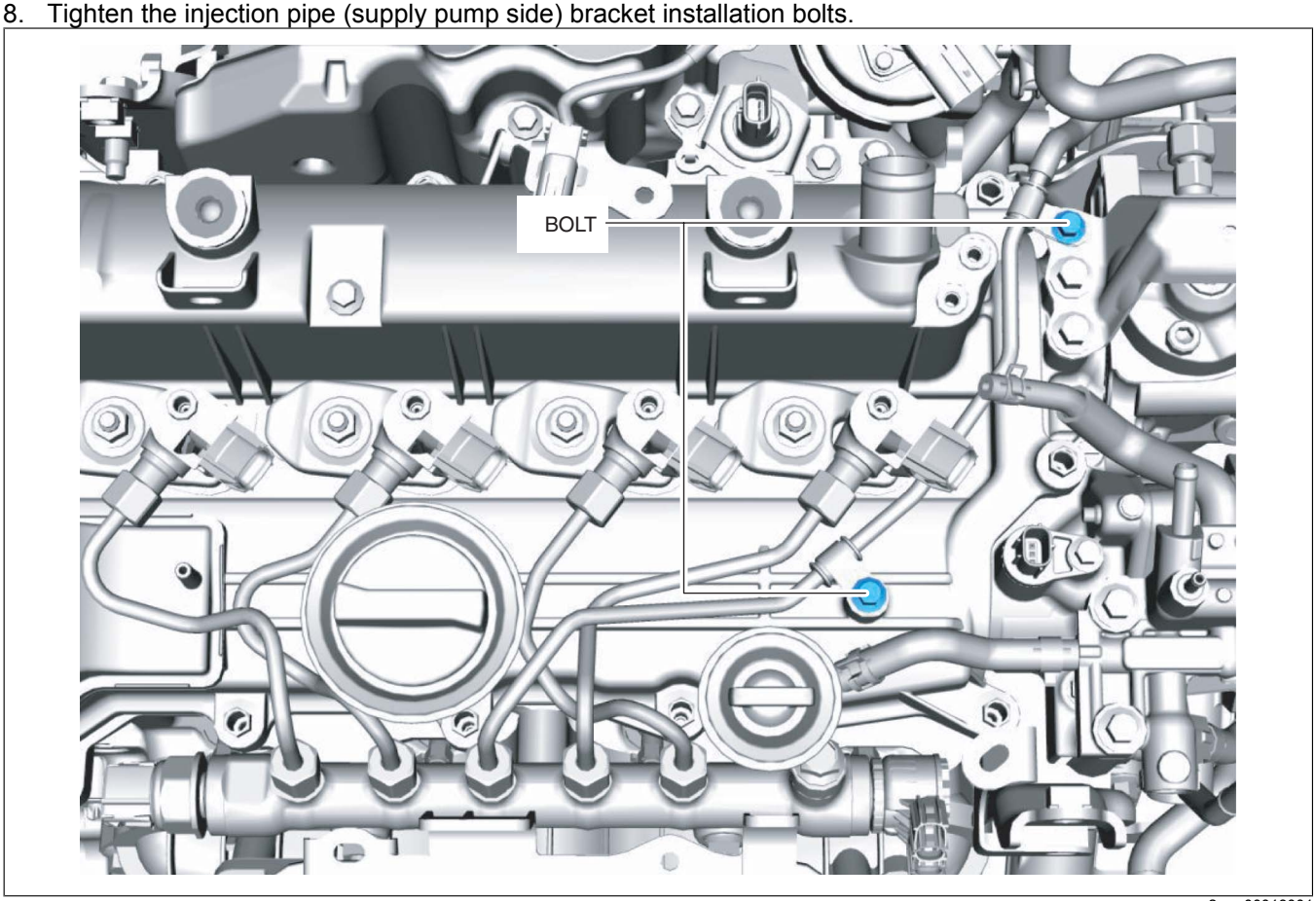
7. Tighten the fuel injector bracket installation nuts in two steps.



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Tightening torque
First pass : 1.5—2.5 N·m {16—25 kgf·cm, 14—22 in·lbf}
Second pass : 117°— 123°

8. Tighten the injection pipe (supply pump side) bracket installation bolts.



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Tightening torque
8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

9. Tighten both ends of the injection pipe (fuel injector side).

Tightening torque
32—42 N·m {3.3—4.2 kgf·m, 24—30 ft·lbf}

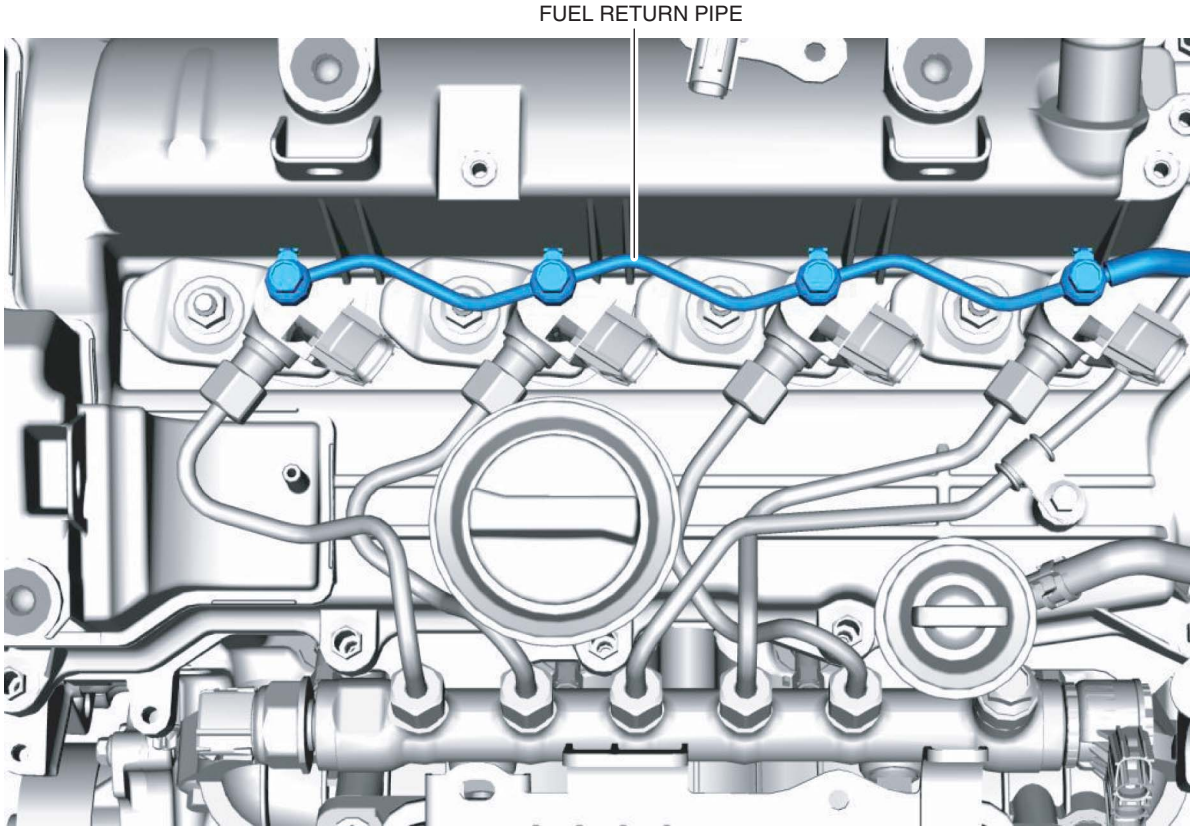
10. Tighten both ends of the injection pipe (supply pump side).

Tightening torque
32—42 N·m {3.3—4.2 kgf·m, 24—30 ft·lbf}

11. Tighten the common rail.

Tightening torque
20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}

12. Tighten the fuel return pipe.

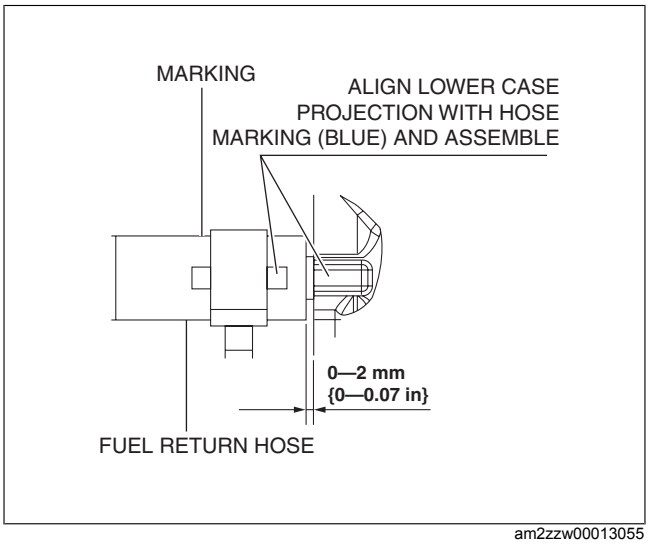


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Tightening torque
8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

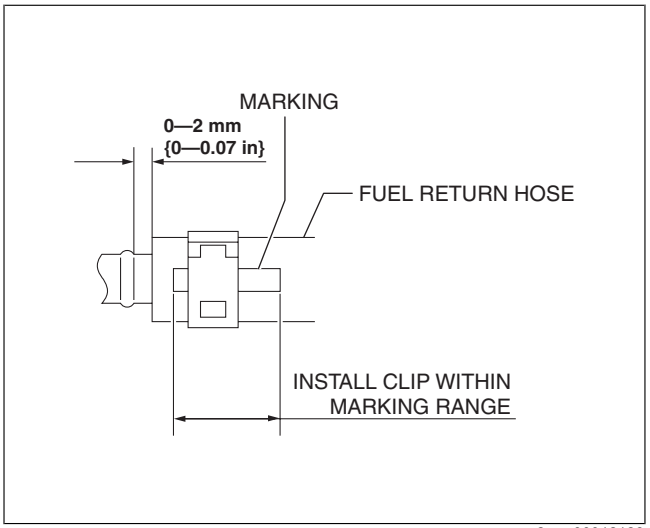
Fuel Return Hose Installation Note
1. Install fuel return hose as shown in the figure.

Lower case side



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Common rail side



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